

Лабораторная работа №6

Bibliography Management in LaTeX

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Цель работы

Освоение использования **natbib**, **BibTeX**, **biblatex** и **Biber**, изучение порядка компиляции и анализ различий между системами управления библиографией.

Работа с natbib и BibTeX

Первый проход pdflatex

- компилятор сообщает об отсутствующих ссылках,
- предупреждения об `undefined citations`,
- PDF создаётся, но ссылки не определены.

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> .\exercise-6-bibliography.pdf
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> notepad exercise-6-bibliography.tex
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> pdflatex exercise-6-bibliography.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise-6-bibliography.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/natbib/natbib.sty)
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdfTeX.def)
(./exercise-6-bibliography.aux)

Package natbib Warning: Citation `Graham1995' on page 1 undefined on input line
11.

Package natbib Warning: Citation `Thomas2008' on page 1 undefined on input line
11.

Package natbib Warning: Citation `Graham1995' on page 1 undefined on input line
13.

Package natbib Warning: Citation `Thomas2008' on page 1 undefined on input line
```

- обработка файла `.aux`,
- подключение базы `references.bib`,
- создание `.bbl`.

```
[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}}]
(./exercise-6-bibliography.aux) ) <c:/Users/ousma/.texlive2025/texmf-var/fonts/
pk/ljfour/jknappen/ec/ecbx1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fo
nts/pk/ljfour/jknappen/ec/ecrm1000.600pk> <c:/Users/ousma/.texlive2025/texmf-va
r/fonts/pk/ljfour/jknappen/ec/ecbx1200.600pk> <c:/Users/ousma/.texlive2025/texm
f-var/fonts/pk/ljfour/jknappen/ec/ecbx1440.600pk>
Output written on exercise-6-bibliography.pdf (1 page, 23795 bytes).
Transcript written on exercise-6-bibliography.log.
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> bibtex exercise-6-bibliography
This is BibTeX, Version 0.99d (TeX Live 2025)
The top-level auxiliary file: exercise-6-bibliography.aux
The style file: plainnat.bst
Database file #1: references.bib
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises>
```

Рис. 2: Подключение BibTeX

Последующие проходы pdflatex

- первый — всё ещё предупреждения,
- второй — ссылки разрешены,
- третий — чистая компиляция без ошибок.

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> pdflatex exercise-6-bibliography.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
 entering extended mode
 (./exercise-6-bibliography.tex
 LaTeX2e <2025-06-01> patch level 1
 L3 programming layer <2025-09-02>
 (c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
 Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
 (c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo)
 (c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
 (c:/texlive/2025/texmf-dist/tex/latex/natbib/natbib.sty)
 (c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
 (./exercise-6-bibliography.aux) (./exercise-6-bibliography.bbl)
 [1(c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map)]
 (./exercise-6-bibliography.aux) ) <c:/Users/ousma/.texlive2025/texmf-var/fonts/
 pk/ljfour/jknappen/ec/ecti1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fo
 nts/pk/ljfour/jknappen/ec/ecrm1000.600pk> <c:/Users/ousma/.texlive2025/texmf-va
 r/fonts/pk/ljfour/jknappen/ec/ecbx1200.600pk> <c:/Users/ousma/.texlive2025/texm
 f-var/fonts/pk/ljfour/jknappen/ec/ecbx1440.600pk>
 Output written on exercise-6-bibliography.pdf (1 page, 38646 bytes).
 Transcript written on exercise-6-bibliography.log.
 PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises>
```

Рис. 3: Финальная компиляция natbib

- корректное отображение цитат,
- список литературы сформирован автоматически.

Exercise 6 - Working with Bibliography

Method 1: Using natbib with BibTeX

The mathematics showcase is from Graham et al. [1995], whereas there is some chemistry in Thomas et al. [2008].

Some parenthetical citations: [Graham et al., 1995] and then [Thomas et al., 2008, p. 56].

[See Graham et al., 1995, pp. 45–48]

Together [Graham et al., 1995, Thomas et al., 2008]

References

Ronald L. Graham, Donald E. Knuth, and Oren Patashnik. *Concrete Mathematics*. Addison-Wesley, 1995.

Christine M. Thomas, Tianbiao Liu, Michael B. Hall, and Marcetta Y. Darensbourg. Series of mixed valent Fe(II)Fe(I) complexes that model the H(OX) state of [FeFe]hydrogenase: Redox properties, density-functional theory investigation, and reactivity with extrinsic CO. *Inorg. Chem.*, 47(15):7009–7024, 2008. doi: 10.1021/ic800654a.

Установка и использование biblatex + Biber

Команды:

- `tlmgr update --self`
- `tlmgr install biblatex biber`

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> kpsewhich biblatex.sty
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> tlmgr update --self
tlmgr.pl: package repository https://mirror.truenetwork.ru/CTAN/systems/texlive/tlnet (not verified: gpg unavailable)
tlmgr.pl: saving backups to C:/texlive/2025/tlpkg/backups
tlmgr.pl: no self-updates for tlmgr available
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> tlmgr install biblatex biber
tlmgr.pl: package repository https://mirror.truenetwork.ru/CTAN/systems/texlive/tlnet (not verified: gpg unavailable)
[1/4, ??:??/??:??] install: biber.windows [23825k]
[2/4, 00:09/00:12] install: biber [1229k]
[3/4, 00:14/00:18] install: biblatex [7428k]
[4/4, 00:21/00:21] install: logreq [7k]
running mktexlsr ...
done running mktexlsr.
tlmgr.pl: package log updated: C:/texlive/2025/texmf-var/web2c/tlmgr.log
tlmgr.pl: command log updated: C:/texlive/2025/texmf-var/web2c/tlmgr-commands.log
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> kpsewhich biblatex.sty
c:/texlive/2025/texmf-dist/tex/latex/biblatex/biblatex.sty
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> notepad exercise-6-complete.tex
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> notepad references.bib
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> pdflatex exercise-6-complete.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise-6-complete.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
```

1. pdflatex
2. biber
3. pdflatex

```
Transcript written on exercise-6-complete.log.  
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> biber exercise-6-complete  
INFO - This is Biber 2.21  
INFO - Logfile is 'exercise-6-complete.blg'  
INFO - Reading 'exercise-6-complete.bcf'  
INFO - Found 2 citekeys in bib section 0  
INFO - Processing section 0  
INFO - Looking for bibtex file 'references.bib' for section 0  
INFO - LaTeX decoding ...  
INFO - Found BibTeX data source 'references.bib'  
INFO - Overriding locale 'en-US' defaults 'normalization = NFD' with 'normalization = prenormalized'  
INFO - Overriding locale 'en-US' defaults 'variable = shifted' with 'variable = non-ignorable'  
INFO - Sorting list 'nyt/global/global/global' of type 'entry' with template 'nyt' and locale 'en-US'  
INFO - No sort tailoring available for locale 'en-US'  
INFO - Writing 'exercise-6-complete.bbl' with encoding 'UTF-8'  
INFO - Output to exercise-6-complete.bbl  
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises>
```

Рис. 6: Запуск biber

- корректная обработка библиографии,
- красивое форматирование,
- автоматическое создание списка литературы.

Exercise 6 - Complete Bibliography Practice

6.1 Reference databases

Created `references.bib` with:

- Article: Thomas et al. (2008) - Chemistry
- Book: Graham et al. (1995) - Mathematics

6.3 The BibTeX workflow with natbib

(For comparison) BibTeX uses commands like:

`\citet{key}` and `\citep{key}`

6.4 The biblatex workflow

Now using `biblatex`:

Textual citation: Graham, Knuth, and Patashnik (1995)

Parenthetical citation: (Thomas et al. 2008)

Cite title only: *Concrete Mathematics*

Auto citation: (Graham, Knuth, and Patashnik 1995)

With page number: (Thomas et al. 2008, p. 56)

With note and pages: (See Graham, Knuth, and Patashnik 1995, pp. 45–48)

Multiple citations: (Graham, Knuth, and Patashnik 1995; Thomas et al. 2008)

6.5 Choosing between BibTeX and BibLaTeX

BibTeX Traditional method, uses `.bst` files, established in many journals

BibLaTeX Modern method, better Unicode support, easier customization

6.7 Hyperlinks

Load `hyperref` package to make citations clickable

Дополнительная компиляция biblatex

LaTeX корректно загрузил **biblatex**, сформировал структуру документа.

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> pdflatex exercise-6-complete.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise-6-complete.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/biblatex/biblatex.sty
(c:/texlive/2025/texmf-dist/tex/generic/pdftexcmds/pdftexcmds.sty
(c:/texlive/2025/texmf-dist/tex/generic/infwarerr/infwarerr.sty)
(c:/texlive/2025/texmf-dist/tex/generic/iftex/iftex.sty)
(c:/texlive/2025/texmf-dist/tex/generic/ltxcms/ltxcms.sty))
```

Рис. 8: Первый проход biblatex

- сравнение методов,
- разные стили цитирования,
- автогенерация списка литературы.

References

- Graham, Ronald L., Donald E. Knuth, and Oren Patashnik (1995). *Concrete Mathematics*. Addison-Wesley.
- Thomas, Christine M. et al. (2008). "Series of Mixed Valent Fe(II)Fe(I) Complexes That Model the H(OX) State of [FeFe]Hydrogenase: Redox Properties, Density-Functional Theory Investigation, and Reactivity with Extrinsic CO". In: *Inorg. Chem.* 47.15, pp. 7009–7024. doi: 10.1021/ic800654a.

Оба метода в одном документе

Работа с файлом exercise-6-both-methods.tex

- запуск BibTeX для natbib-блока,
- компиляция pdflatex два раза,
- обработка смешанного документа.

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> .\exercise-6-complete.pdf
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> notepad exercise-6-both-methods.tex
PS C:\Users\ousma\Documents\latex-exercises\chapter-5-exercises> pdflatex exercise-6-both-methods.tex
>> bibtex exercise-6-both-methods
>> pdflatex exercise-6-both-methods.tex
>> pdflatex exercise-6-both-methods.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise-6-both-methods.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo)
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/natbib/natbib.sty)
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdfTeX.def)
No file exercise-6-both-methods.aux.

Package natbib Warning: Citation `Graham1995' on page 1 undefined on input line
15.

Package natbib Warning: Citation `Thomas2008' on page 1 undefined on input line
17.
```

Рис. 10: Компиляция смешанного документа

Корректное отображение:

- текстовых цитат,
- скобочных,
- множественных.

Exercise 6 - Both Bibliography Methods

Part A: BibTeX workflow (traditional)

Using `natbib` package:

Textual: Graham et al. [1995]

Parenthetical: [Thomas et al., 2008]

Multiple: [Graham et al., 1995, Thomas et al., 2008]

References

Ronald L. Graham, Donald E. Knuth, and Oren Patashnik. *Concrete Mathematics*. Addison-Wesley, 1995.

Christine M. Thomas, Tianbiao Liu, Michael B. Hall, and Marcetta Y. Darensbourg. Series of mixed valent Fe(II)Fe(I) complexes that model the H(OX) state of [FeFe]hydrogenase: Redox properties, density-functional theory investigation, and reactivity with extrinsic CO. *Inorg. Chem.*, 47(15):7009–7024, 2008. doi: 10.1021/ic800654a.

Итоги работы

В ходе работы были освоены:

- применение двух систем библиографии: **BibTeX/natbib** и **BibLaTeX/Biber**;
- порядок компиляции для каждой системы;
- создание и подключение собственных записей в **.bib**;
- анализ поведения при отсутствии ключей;
- получение корректного списка литературы через обе системы.

Обе методики успешно работают и позволяют гибко оформлять научные документы.